



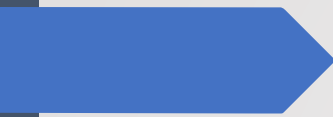
# **BSM422 Bilgisayarla Görme ve Görüntüleme Teknikleri**

**Dr. Öğr. Üyesi Mehmet Zahid YILDIRIM**

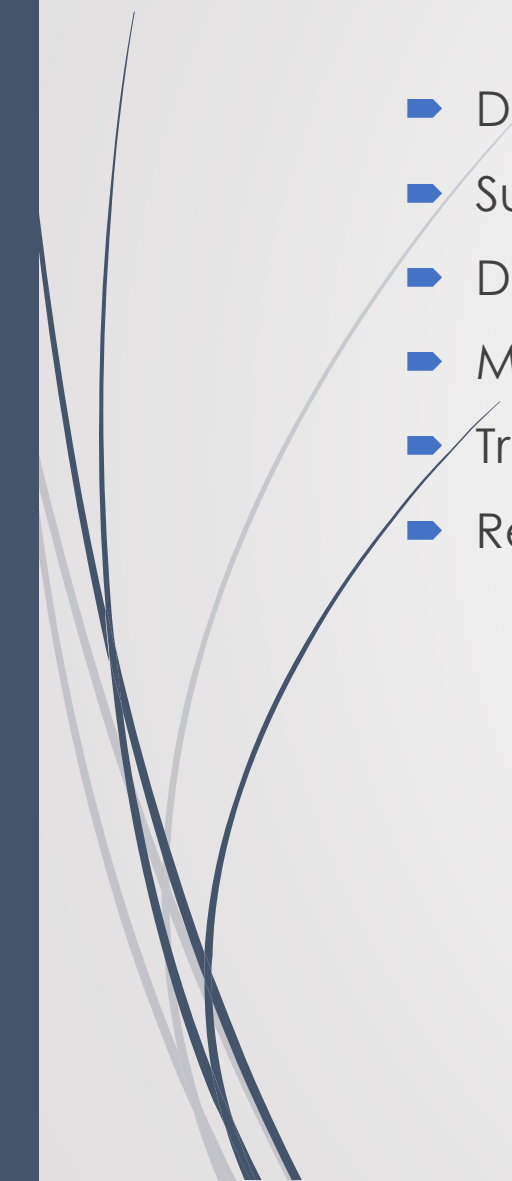


# Bu ders;

- Veri Artırım Teknikleri Üzerine Yapılmış Bir Çalışma
  - Baştan Sona Proje Adımları
  - Kullanılan Yöntem ve Metriklerin Analizi
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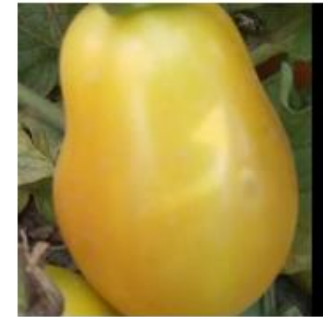
# Proje Geliştirme Adımları

- Dataset (Verisetinin Hazırlanması)
  - Subset (Alt Veriseti Oluşturma)
  - Data Augmentation (Veri Artırım Stratejisi)
  - Models and Metrics (Kullanılan Modeller ve Metrikler)
  - Train – Test (Eğitim ve Test Süreci)
  - Results (Deneysel Sonuçların Yorumlanması)
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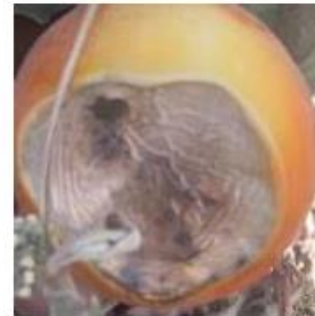
# Dataset

- ▶ TOM2024 plant disease dataset (*Data in Brief*) – 2025
- ▶ Tomato subset selected (**Tomato**, Onion, Maize)
- ▶ Binary classification:
  - Healthy, Unhealthy
- ▶ Dataset size:
  - 1334 healthy
  - 2818 unhealthy
- ▶ Class imbalance problem
  - Healthy samples: 1334 → 2668

Healthy

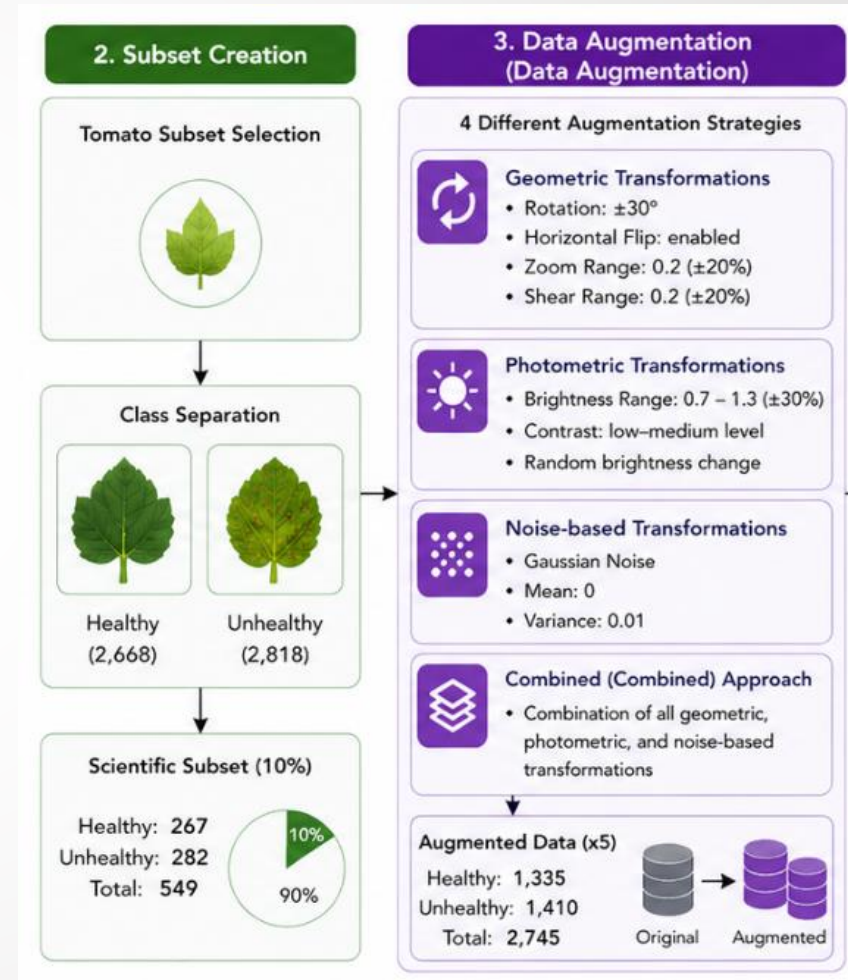


Unhealthy



# Data Augmentation Strategies

- Scientifically selected subset (10%)
  - 267 healthy / 282 unhealthy
- Stratified sampling approach
- Simulates limited data scenario
- Data augmentation applied ( $\times 5$ )
- Four augmentation strategies:
  - Geometric
  - Photometric
  - Noise
  - Combined

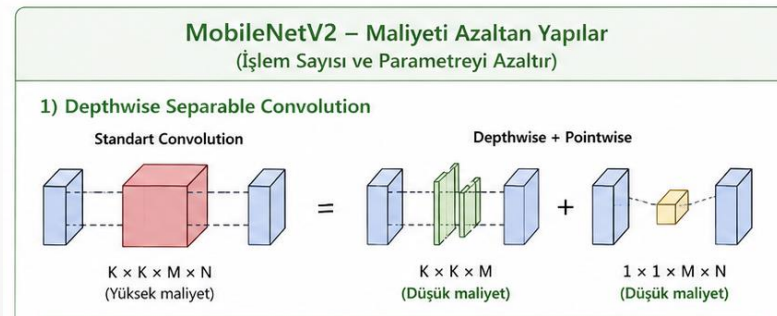
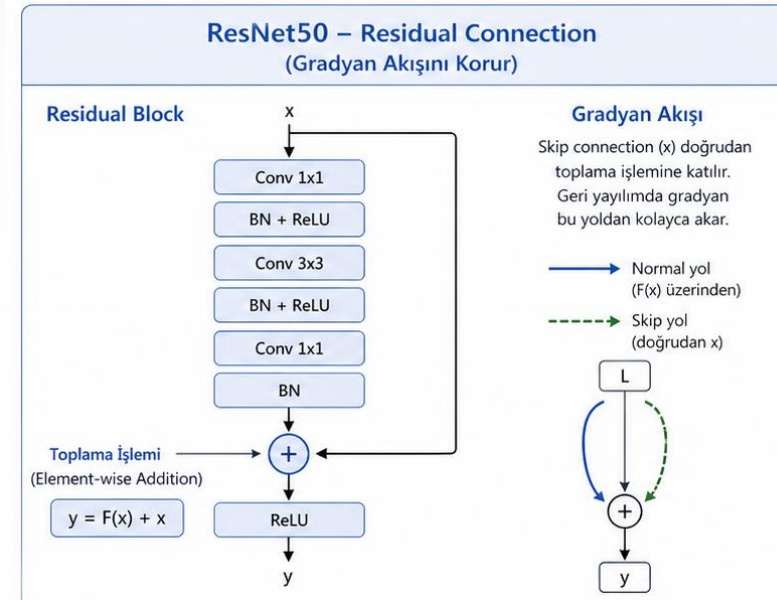


# Data Size for All Subsets

| <b>Data Status</b>    | <b>Healty Samples</b> | <b>Unhealthy Samples</b> | <b>Total Samples</b> |
|-----------------------|-----------------------|--------------------------|----------------------|
| Original Data Set     | 2668                  | 2818                     | 5486                 |
| Subset reduced to 10% | 267                   | 282                      | 549                  |
| 5x augmented dataset  | 1335                  | 1410                     | 2745                 |

# Deep Learning Models Used

- Transfer Learning approach
- Pre-trained CNN models
- ResNet50
  - Deep architecture
  - Residual connections
- MobileNetV2
  - Lightweight model
  - Efficient computation



# Evaluation Metrics

## ➤ Accuracy

- Modelin toplamda ne kadar doğru tahmin yaptığını gösterir. Yani doğru tahminlerin tüm tahminlere oranıdır.

## ➤ Precision

- Precision, modelin “pozitif” dediği örneklerin ne kadarının gerçekten doğru olduğunu ölçer.

## ➤ Recall

- Recall, gerçek pozitif örneklerin ne kadarını modelin doğru yakalayabildiğini gösterir.

## ➤ F1-score

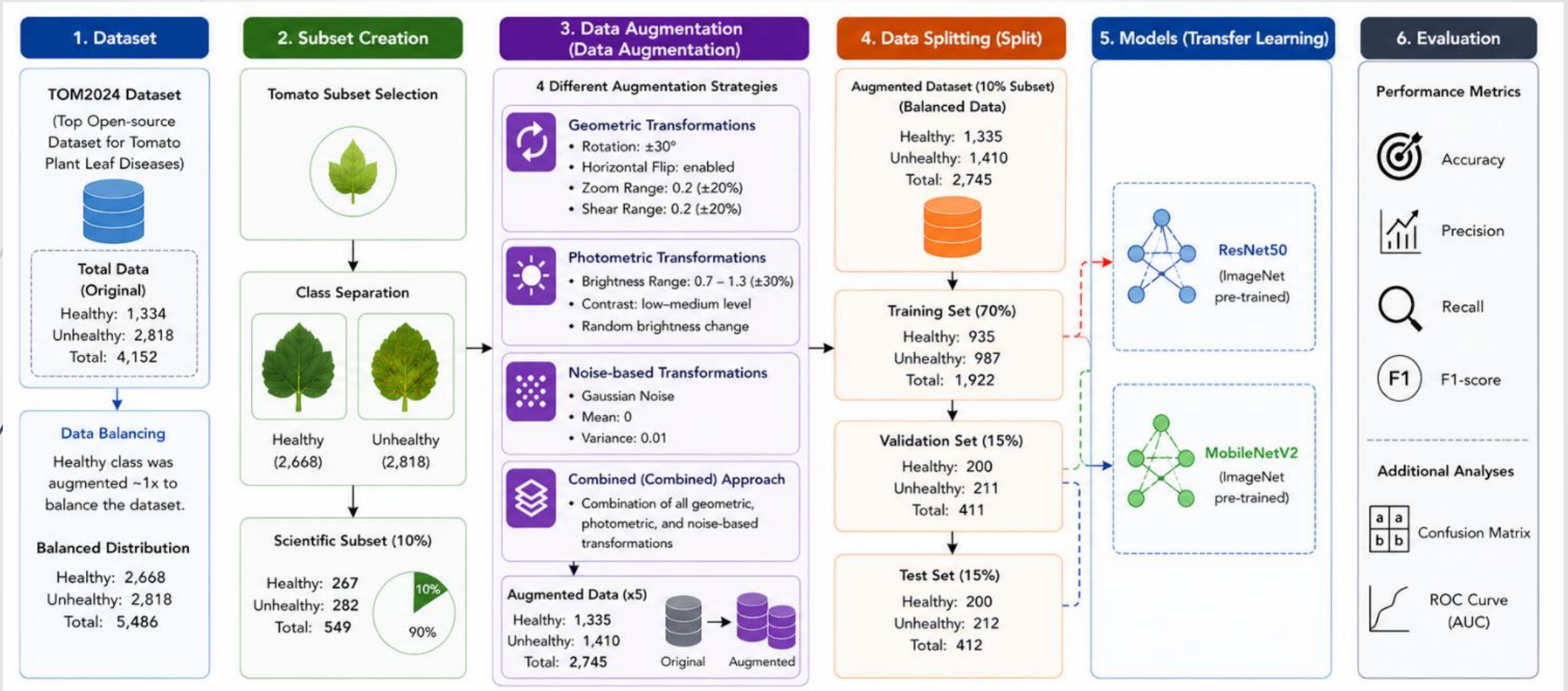
- F1-score, precision ve recall'un dengeli bir birleşimidir.

## ➤ Loss Curve (Acc/Val)

## ➤ Confusion Matrix

## ➤ ROC Curve

# System Architecture



# Experimental Results

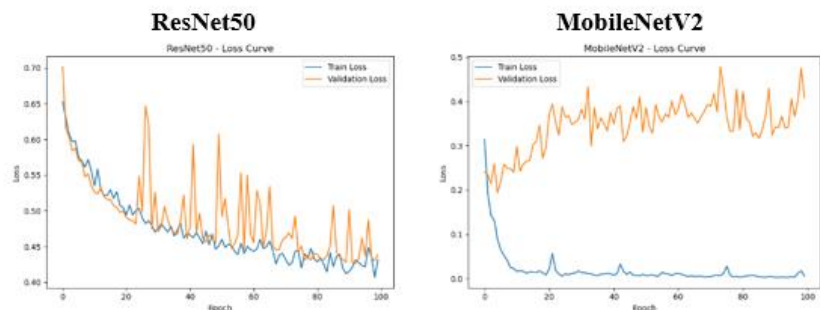
## Analysis of Evaluation Metrics

| Dataset                  | Model       | Accuracy | Precision | Recall | F1-Score |
|--------------------------|-------------|----------|-----------|--------|----------|
| All Data                 | ResNet50    | 0.7939   | 0.7400    | 0.9220 | 0.8211   |
|                          | MobileNetV2 | 0.9297   | 0.9176    | 0.9480 | 0.9326   |
| 10% Subset               | ResNet50    | 0.6463   | 0.8824    | 0.3571 | 0.5085   |
|                          | MobileNetV2 | 0.8659   | 0.8780    | 0.8571 | 0.8675   |
| Geometric Augmentation   | ResNet50    | 0.7463   | 0.7548    | 0.7476 | 0.7512   |
|                          | MobileNetV2 | 0.9171   | 0.9112    | 0.9286 | 0.9198   |
| Photometric Augmentation | ResNet50    | 0.7512   | 0.6862    | 0.9476 | 0.7960   |
|                          | MobileNetV2 | 0.8951   | 0.8957    | 0.9000 | 0.8979   |
| Noise Augmentation       | ResNet50    | 0.7805   | 0.7885    | 0.7810 | 0.7847   |
|                          | MobileNetV2 | 0.8683   | 0.9105    | 0.8238 | 0.8650   |
| Combined Augmentation    | ResNet50    | 0.7439   | 0.7442    | 0.7619 | 0.7529   |
|                          | MobileNetV2 | 0.8854   | 0.8937    | 0.8810 | 0.8873   |

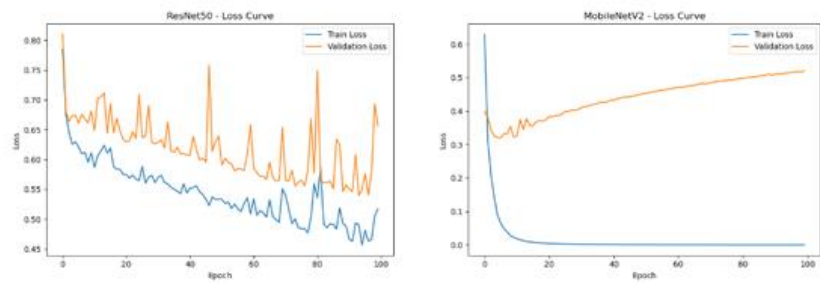
# Experimental Results

## Analysis of Loss Curves

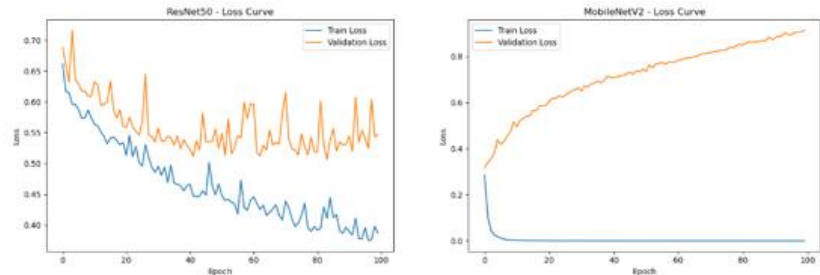
All Data



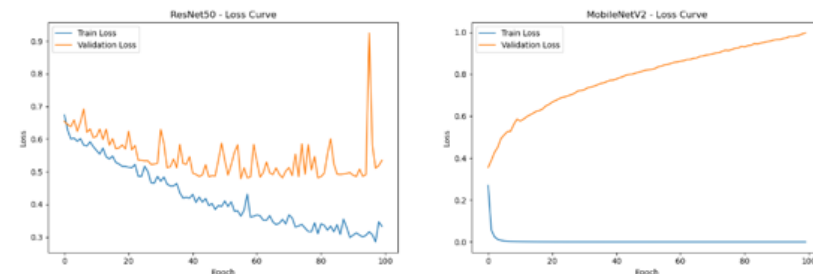
10% Subset



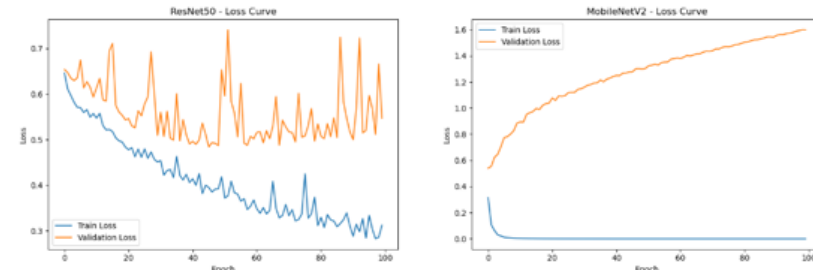
Geometric Augmentation



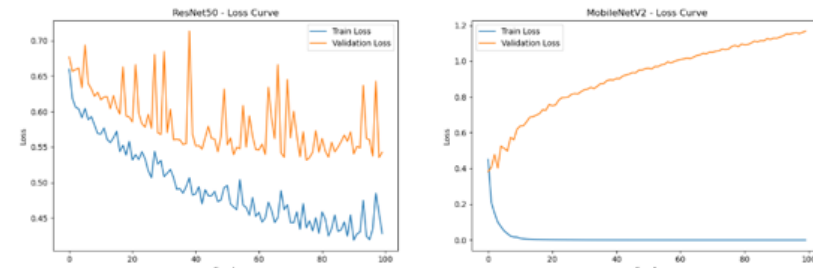
Photometric Augmentation



Noise Augmentation



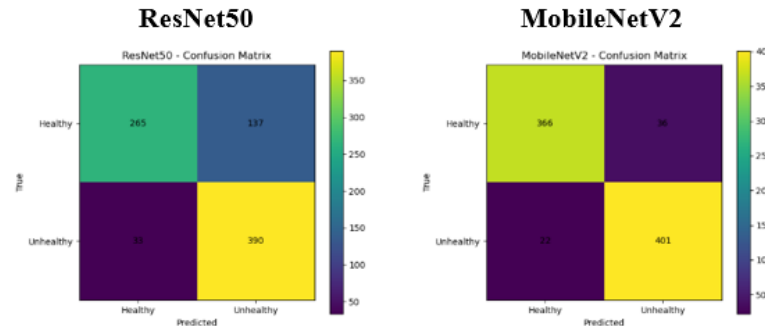
Combined Augmentation



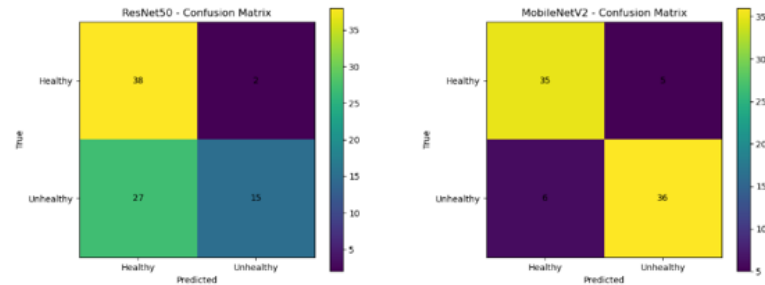
# Experimental Results

## Analysis of Confusion Matrices

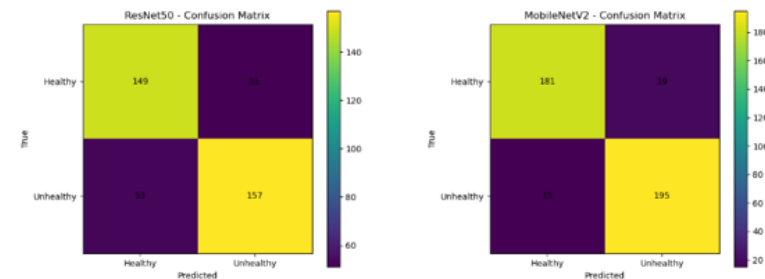
All Data



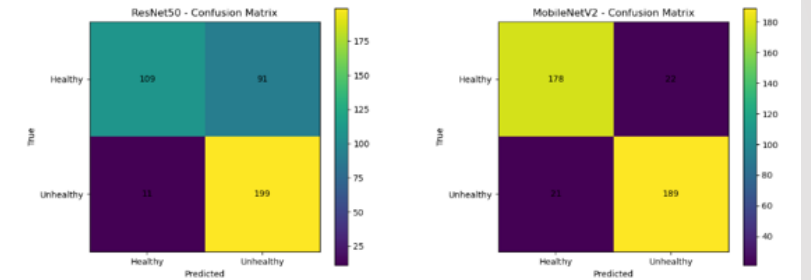
10% Subset



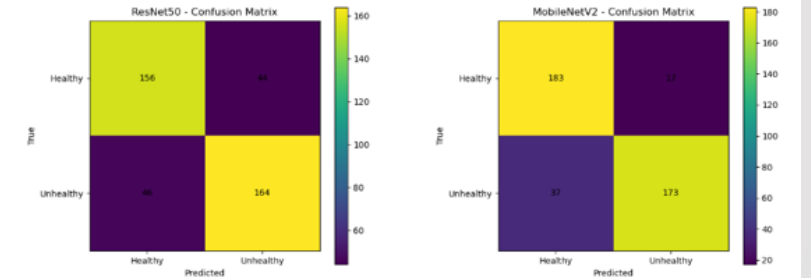
Geometric Augmentation



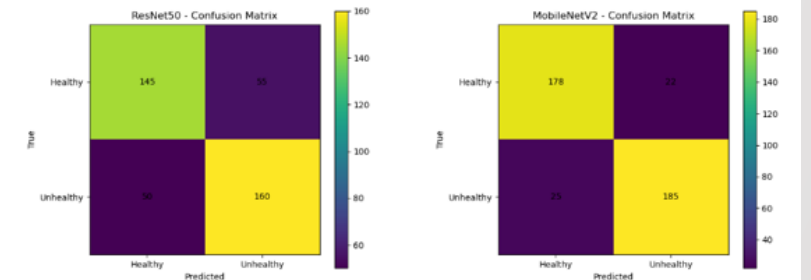
Photometric Augmentation



Noise Augmentation



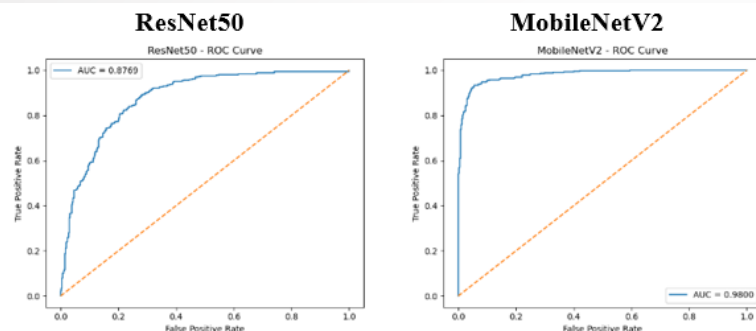
Combined Augmentation



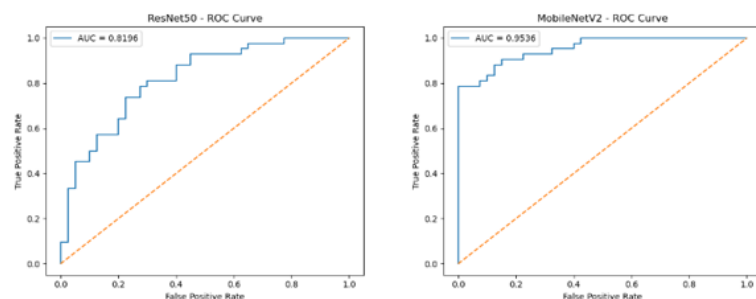
# Experimental Results

## Analysis of ROC Curves

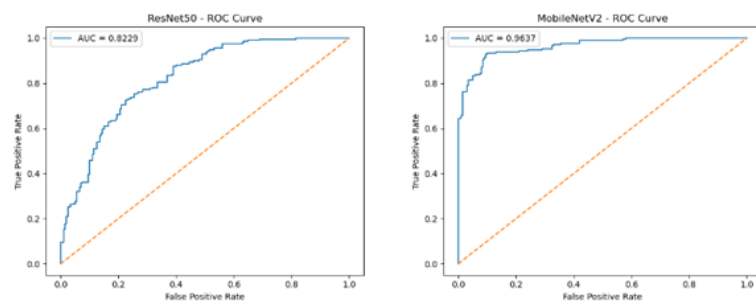
All Data



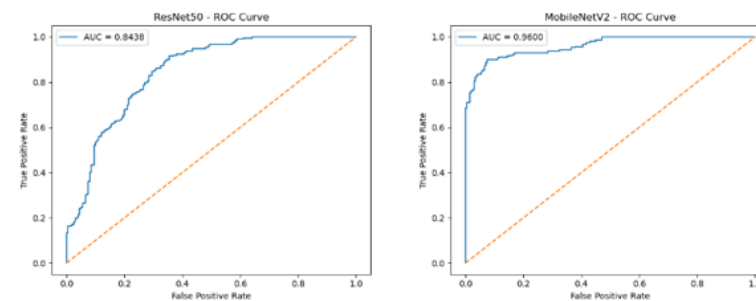
10% Subset



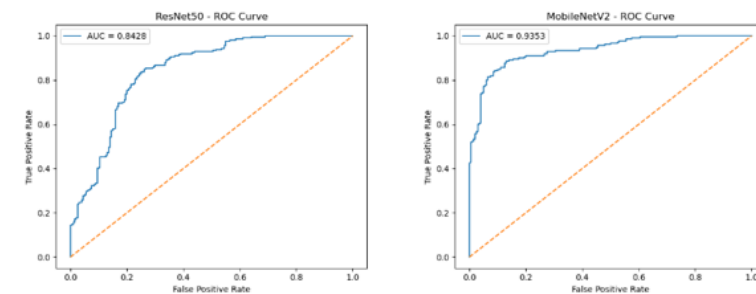
Geometric Augmentation



Photometric Augmentation



Noise Augmentation



Combined Augmentation

